

Numerical Analysis 1 Midterm Exam

1. a. Use bisection method to find 4 interval for the following equation on $[0,1]$
 $2x + 3 \cos(\pi x) - e^x = 0$
 b. How many bisections are needed to achieve 10^{-10} accuracy
2. Use inverse interpolation method to find an approximation to the solution $e^x - x = 0$, using the data (find x value for $f(x)=0$.)

x	0.3	0.4	0.5	0.6
$f(x)$	0.740	0.670	0.606	0.548

Note : Take 3 digits after the point.

3. Generate the divided difference table for the following data and approximate $f(0.05)$.

x	0.0	0.2	0.4	0.6	0.8
$f(x)$	1.00000	1.22140	1.49182	1.82212	2.22554

Note : Take 5 digits after the point.

4. Solve the following equation system by LU Method.

$$\begin{aligned}
 x_1 - x_2 &= 2, \\
 2x_1 + 2x_2 + 3x_3 &= -1, \\
 -x_1 + 3x_2 + 2x_3 &= 4.
 \end{aligned}$$